Shashank Gugnani

Senior Engineering Manager | Database Systems | Distributed Storage

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Profile

Senior engineering leader with expertise in database systems, storage, and developer platforms. Currently leading a team at Oracle delivering next-generation application development technologies for Oracle Database, including JSON-Relational Duality, developer APIs, and GraphQL integration. Proven track record of transitioning research innovations into production database features at global scale. Author of 20+ peer-reviewed publications and inventor on multiple database patents.

Leadership & Technical Focus

- Engineering leadership for database and platform teams
- Developer-facing database abstractions and APIs
- JSON, relational, and hybrid data models
- Distributed systems, storage, and persistent memory
- Research-to-production technology transfer

Professional Experience

Oracle | Redwood Shores, CA 2021 – Present

Senior Engineering Manager

Leadership & scope

- Lead and grow a team of engineers building Agentic AI and App Devolvement technologies inside the Oracle Database.
- Own technical strategy and delivery for multi-release road-map initiatives spanning the database kernel and developer tooling.
- Partner cross-functionally with product management, database kernel, documentation, and support teams.
- Mentor senior engineers and interns; support promotions and career growth.

Selected products & initiatives

- **JSON-Relational Duality:** Led core engineering efforts enabling a unified programming model across JSON and relational data with transactional guarantees.
- **JSON-to-Duality Migrator:** Designed and delivered automated tooling to migrate document-centric schemas and workloads to Oracle duality views.
- **Oracle Database Support for GraphQL:** Led database-native GraphQL integration, aligning API semantics with Oracle Database security, performance, and scalability requirements.
- **DBMS_DEVELOPER:** Contributed to developer-oriented PL/SQL APIs improving productivity, extensibility, and observability for database developers.

Individual contributions

- Designed and implemented DML and transaction-layer components for JSON-Relational Duality.
- Co-authored publications and filed database systems patents.

Education

The Ohio State University | Columbus, OH 2015 – 2021
Ph.D., Computer Science
Thesis: *Designing Scalable Storage Systems for Non-Volatile Memory*

BITS Pilani | India 2011 – 2015
B.E., Computer Science
Thesis: *Evaluation and Implementation of Hadoop in a Scientific Gateway Environment*

Selected Patents

1. **S. Gugnani**, A. Mylavarapu, S. Pendse, T. Lahiri, S. Ahmed, S. Gowda, B. Hammerschmidt, and Z. Liu, "Automated migration from a document database to a relational database," US Patent 20250147936A1.
2. Z. Liu, J. Loaiza, S. Abraham, S. Bose, H. Chang, **S. Gugnani**, B. Hammerschmidt, T. Lahiri, Y. Lu, D. McMahon, A. Mishra, A. Mylavarapu, S. Pendse, and A. Raghavan, "Natively supporting JSON duality view in a database management system," US Patent 12287777B2.
3. **S. Gugnani**, S. Guthridge, F. Schmuck, T. Anderson, and D. Bhagwat, "Fine-Grained Forced Cache Eviction," US Patent 201909738.

Selected Publications

1. **S. Gugnani**, S. Ahmed, S. Pednekar, S. Gowda, S. Tandon, and S. Pendse, "From JSON to Duality: Automated Application Migration from Document to Relational Databases," SIGMOD 2026. [DOI]
2. **S. Gugnani**, Z. Liu, H. Chang, B. Hammerschmidt, S. Kareenhalli, K. Kumar, T. Lahiri, Y. Lu, D. McMahon, A. Mylavarapu, S. Pendse, and A. Raghavan, "JSON Relational Duality: A Revolutionary Combination of Document, Object, and Relational Models," SIGMOD 2025. [DOI]
3. **S. Gugnani** and X. Lu, "DStore: A Fast, Tailless, and Quiescent-Free Object Store for PMEM," HPDC 2021. [DOI]
4. **S. Gugnani**, A. Kashyap, and X. Lu, "Understanding the Idiosyncrasies of Real Persistent Memory," VLDB 2021. [DOI]
5. **S. Gugnani**, T. Li, and X. Lu, "NVMe-CR: A Scalable Ephemeral Storage Runtime for Checkpoint/Restart with NVMe-over-Fabrics," IPDPS 2021. [DOI]

Mentorship, Service & Recognition

- Mentored engineers, interns, and Ph.D. students across research and production teams; supported intern-to-full-time conversions and internal promotions.
- Reviewer for SIGMOD'26, IEEE TPDS, MASCOTS'19, IPDPS'18, and ICS'17.
- Speaker at SIGMOD, VLDB, HPDC, IPDPS, SC, IISWC, and GraphQLConf.
- Guest lectured at UC Merced, UC Berkeley, and UC San Diego.
- Presented at the ACM Student Research Competition (SC 2017, 2018).

Technical Skills

Languages: C, C++, Java, SQL, Bash
Domains: Database systems, storage engines, developer platforms